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T12

SUB : ASD&D

EXPERIMENT – 2

AIM : To Containerize Application using Docker.

LO Mapped : LO4

LO4 : Containerize , orchestrate , and deploy applications using Docker , Docker Compose , Kubernetes and Cloud Infrastructure .

THEORY :

1. Introduction to Docker

Docker is an open-source platform used to develop, package, deploy, and run applications in a standardized environment called a container. It uses operating-system-level virtualization to isolate applications and their dependencies without requiring a complete virtual machine.

Docker solves the common problem of "it works on my machine" by packaging an application along with its required libraries, dependencies, and configuration. This allows the application to run consistently across different environments.

2. Containerization

Containerization is the process of packaging an application and all its required dependencies into a lightweight, portable container.

Containers share the host operating system's kernel, making them generally faster and more lightweight than traditional virtual machines.

Advantages of containerization include:

- Portability

- Fast startup time
- Lightweight resource usage
- Application isolation
- Easy deployment
- Consistent environments
- Scalability

3. Docker Image

A Docker image is a read-only, immutable template that contains everything required to run an application. It includes:

- Application source code
- Runtime environment
- Libraries
- Dependencies
- Environment configuration
- Instructions for starting the application

Images are commonly created using a Dockerfile. An image can be used to create multiple containers.

For this experiment, the Docker image was named:

aryank

4. Docker Container

A Docker container is a running or stopped instance of a Docker image. It provides an isolated environment where the application executes.

A container is created from an image using the docker run command. Multiple containers can be created from the same image, and each container can have its own name, network configuration, and runtime state.

In this experiment, the container was named:

aryank-container

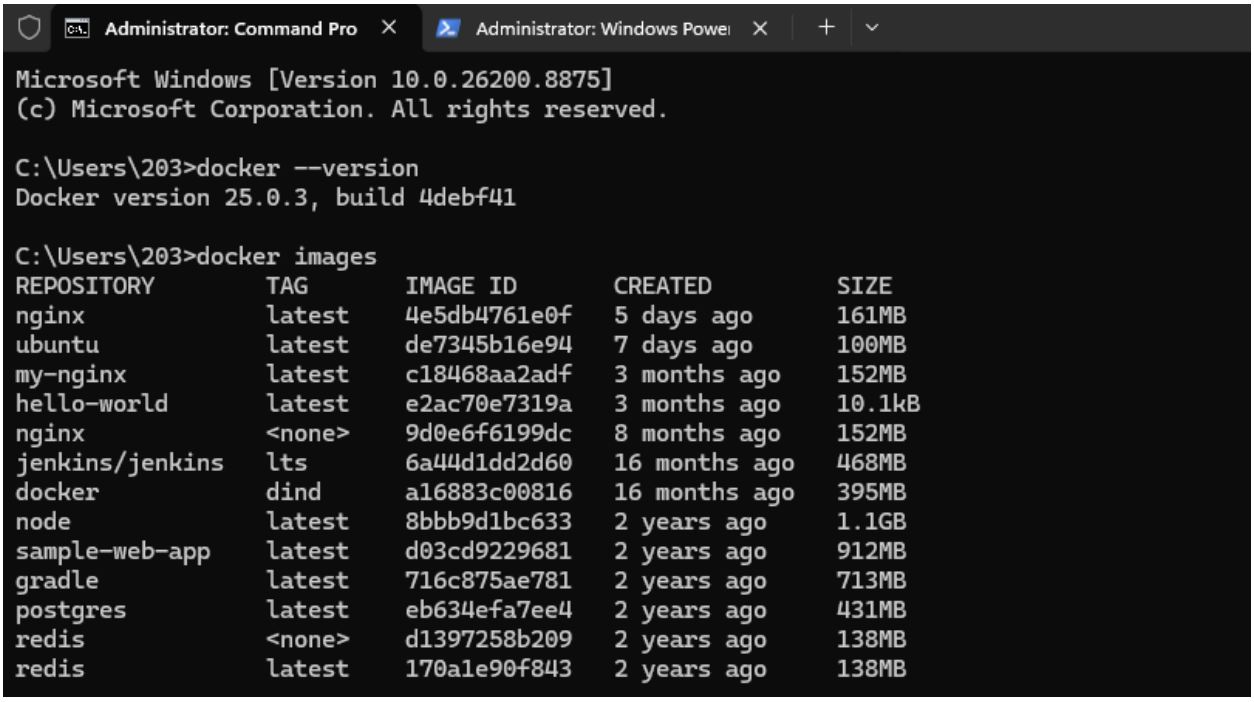
and a background container was named:

aryank-background

5. Difference Between Docker Image and Container

Feature	**Docker Image**	**Docker Container**
Definition	Template used to create containers	Instance of an image
State	Immutable/read-only	Has a runtime state
Purpose	Stores application and dependencies	Runs the application
Creation	Built using a Dockerfile	Created using docker run
Example	aryank	aryank-container
Multiple Instances	One image can create many containers	Each container is an individual instance
Lifecycle	Exists until deleted	Can be started, stopped, and removed

OUTPUT :



```
Administrator: Command Pro    Administrator: Windows Powe

Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\203>docker --version
Docker version 25.0.3, build 4debf41

C:\Users\203>docker images
REPOSITORY        TAG       IMAGE ID       CREATED         SIZE
nginx             latest    4e5db4761e0f   5 days ago      161MB
ubuntu            latest    de7345b16e94   7 days ago      100MB
my-nginx          latest    c18468aa2adf   3 months ago    152MB
hello-world       latest    e2ac70e7319a   3 months ago    10.1kB
nginx             <none>    9d0e6f6199dc   8 months ago    152MB
jenkins/jenkins   lts       6a44d1dd2d60   16 months ago   468MB
docker            dind      a16883c00816   16 months ago   395MB
node              latest    8bbb9d1bc633   2 years ago     1.1GB
sample-web-app    latest    d03cd9229681   2 years ago     912MB
gradle            latest    716c875ae781   2 years ago     713MB
postgres          latest    eb634efa7ee4   2 years ago     431MB
redis             <none>    d1397258b209   2 years ago     138MB
redis             latest    170a1e90f843   2 years ago     138MB
```

```
C:\Users\203>docker search ubuntu
NAME                            DESCRIPTION                                     STARS     OFFICIAL
ubuntu                          Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
ubuntu/squid                    Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx                    Nginx, a high-performance reverse proxy & we…   141
ubuntu/bind9                    BIND 9 is a very flexible, full-featured DNS…   120
ubuntu/kafka                    Apache Kafka, a distributed event streaming …   62
ubuntu/zookeeper                ZooKeeper maintains configuration informatio…   15
ubuntu/apache2                  Apache, a secure & extensible open-source HT…   104
ubuntu/prometheus               Prometheus is a systems and service monitori…   80
ubuntu/jre                      Chiseled Java runtime based on Ubuntu. Long-…   25
ubuntu/dotnet-aspnet            Chiselled Ubuntu runtime image for ASP.NET a…   26
ubuntu/python                   A chiselled Ubuntu rock with the Python runt…   33
ubuntu/mysql                    MySQL open source fast, stable, multi-thread…   74
ubuntu/postgres                 PostgreSQL is an open source object-relation…   45
ubuntu/dotnet-runtime           Chiselled Ubuntu runtime image for .NET apps…   24
ubuntu/redis                    Redis, an open source key-value store. Long-…   26
ubuntu/dotnet-deps              Chiselled Ubuntu for self-contained .NET & A…   16
ubuntu/grafana                  Grafana, a feature rich metrics dashboard & …   12
ubuntu/memcached                Memcached, in-memory keyvalue store for smal…   7
ubuntu/jdk                      Ubuntu rock for OpenJDK                         0
ubuntu/rabbitmq                 Ubuntu-based RabbitMQ image - an open source…   0
ubuntu/mimir                    Ubuntu ROCK for Mimir, a horizontally scalab…   0
ubuntu/loki                     Grafana Loki, a log aggregation system like …   2
ubuntu/traefik                  Ubuntu ROCK for Traefik, a modern HTTP rever…   3
ubuntu/opentelemetry-collector  This is a rock for the OpenTelemetry Collect…   0
ubuntu/alertmanager             Ubuntu Rock for Alertmanager. It handles ale…   1

C:\Users\203>docker search ubuntu --filter STARS=100
NAME             DESCRIPTION                                     STARS     OFFICIAL
ubuntu           Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
ubuntu/squid     Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx     Nginx, a high-performance reverse proxy & we…   141
ubuntu/bind9     BIND 9 is a very flexible, full-featured DNS…   120
ubuntu/apache2   Apache, a secure & extensible open-source HT…   104
```

```
C:\Users\203>docker run ubuntu /bin/bash

C:\Users\203>docker ps -a
CONTAINER ID   IMAGE                COMMAND                  CREATED          STATUS                      PORTS                                NAMES
79d59be7b7ef   ubuntu               "/bin/bash"              10 seconds ago   Exited (0) 8 seconds ago                                         quirky_matsumoto
c8fe146dd791   ubuntu               "bash"                   19 hours ago     Exited (255) 7 minutes ago                                       pedantic_hypatia
2ba1c97abb52   ubuntu               "/bin/bash"              19 hours ago     Exited (127) 19 hours ago                                        suspicious_hugle
0fb51210958c   my-nginx             "/docker-entrypoint.…"   3 months ago     Exited (255) 19 hours ago   0.0.0.0:9091->80/tcp                 thirsty_pare
fe399986e504   9d0e6f6199dc         "/docker-entrypoint.…"   3 months ago     Exited (255) 19 hours ago   0.0.0.0:9090->80/tcp                 infallible_germain
5be8ea3f3bb2   9d0e6f6199dc         "/docker-entrypoint.…"   3 months ago     Created                                                          ecstatic_jennings
1e61b2cb20bf   hello-world          "/hello"                 3 months ago     Exited (0) 3 months ago                                          nervous_elbakyan
23106efee7f4   9d0e6f6199dc         "/docker-entrypoint.…"   8 months ago     Exited (255) 4 months ago   0.0.0.0:8081->80/tcp                 my-nginx
d1c21cfdc92b   jenkins/jenkins:lts  "/usr/bin/tini -- /u…"   16 months ago    Exited (255) 8 months ago   50000/tcp, 0.0.0.0:9090->8080/tcp    jenkins
6d895ae2b5eb   d1397258b209         "docker-entrypoint.s…"   2 years ago      Exited (0) 2 years ago                                           nice_greider
1f99a9cc77bf   gradle               "/__cacert_entrypoin…"   2 years ago      Exited (0) 2 years ago                                           elastic_stonebraker
77e3fc5c920f   node                 "docker-entrypoint.s…"   2 years ago      Exited (0) 2 years ago                                           con234
7b58e9788533   node                 "docker-entrypoint.s…"   2 years ago      Exited (0) 2 years ago                                           containerblah
0dc3412adadd   postgres             "docker-entrypoint.s…"   2 years ago      Exited (1) 2 years ago                                           stoic_mclaren

C:\Users\203>docker run -it ubuntu /bin/bash
root@9b433640960c:/# mkdir demo
root@9b433640960c:/# cd demo
root@9b433640960c:/demo# pwd
/demo
root@9b433640960c:/demo#
exit
```

```
C:\Users\203\Desktop\docker-project>docker build -t aryank .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  4.096kB
Step 1/7 : FROM python:3.12-slim
3.12-slim: Pulling from library/python
062e450697fa: Already exists
98db2485a0d0: Pull complete
48347b15c85f: Pull complete
fd079632edc0: Pull complete
Digest: sha256:57cd7c3a7a273101a6485ba99423ee568157882804b1124b4dd04266317710de
Status: Downloaded newer image for python:3.12-slim
 ---> 25c5b8011a34
Step 2/7 : WORKDIR /app
 ---> Running in 0e6b9fecaddf
 ---> Removed intermediate container 0e6b9fecaddf
 ---> 554abaab7e7f
Step 3/7 : COPY requirements.txt .
 ---> f08d31dc0493
Step 4/7 : RUN pip install --no-cache-dir -r requirements.txt
 ---> Running in ce99646cb8d9
Collecting Flask (from -r requirements.txt (line 1))
  Downloading flask-3.1.3-py3-none-any.whl.metadata (3.2 kB)
Collecting blinker>=1.9.0 (from Flask->-r requirements.txt (line 1))
```

```
C:\Users\203\Desktop\docker-project>docker build -t aryank . --no-cache
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  4.096kB
Step 1/7 : FROM python:3.12-slim
 ---> 25c5b8011a34
Step 2/7 : WORKDIR /app
 ---> Running in 0ce3d0898e4e
 ---> Removed intermediate container 0ce3d0898e4e
 ---> ed8efe307646
Step 3/7 : COPY requirements.txt .
 ---> ee6056b81239
Step 4/7 : RUN pip install --no-cache-dir -r requirements.txt
 ---> Running in f22e49e2e2bc
Collecting Flask (from -r requirements.txt (line 1))
  Downloading flask-3.1.3-py3-none-any.whl.metadata (3.2 kB)
Collecting blinker>=1.9.0 (from Flask->-r requirements.txt (line 1))
  Downloading blinker-1.9.0-py3-none-any.whl.metadata (1.6 kB)
Collecting click>=8.1.3 (from Flask->-r requirements.txt (line 1))
  Downloading click-8.4.2-py3-none-any.whl.metadata (2.6 kB)
Collecting itsdangerous>=2.2.0 (from Flask->-r requirements.txt (line 1))
  Downloading itsdangerous-2.2.0-py3-none-any.whl.metadata (1.9 kB)
Collecting jinja2>=3.1.2 (from Flask->-r requirements.txt (line 1))
  Downloading jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting markupsafe>=2.1.1 (from Flask->-r requirements.txt (line 1))
```

```
Loading personal and system profiles took 2132ms.
(base) PS C:\Users\203> docker ps
CONTAINER ID   IMAGE     COMMAND           CREATED              STATUS             PORTS                    NAMES
103ab71a7225   aryank    "python app.py"   About a minute ago   Up About a minute   0.0.0.0:5000->5000/tcp   aryank-container
(base) PS C:\Users\203> docker ps -a
CONTAINER ID   IMAGE                COMMAND                  CREATED          STATUS                      PORTS                                NAMES
103ab71a7225   aryank               "python app.py"          2 minutes ago    Up 2 minutes                0.0.0.0:5000->5000/tcp               aryank-container
9b4336d0960c   ubuntu               "/bin/bash"              48 minutes ago   Exited (0) 42 minutes ago                                        eager_bartik
79d59be7b7ef   ubuntu               "/bin/bash"              49 minutes ago   Exited (0) 49 minutes ago                                        quirky_matsumoto
c8fe146dd791   ubuntu               "bash"                   20 hours ago     Exited (255) 57 minutes ago                                      pedantic_hypatia
2ba1c97abb52   ubuntu               "/bin/bash"              20 hours ago     Exited (127) 20 hours ago                                        suspicious_hugle
0fb51210958c   my-nginx             "/docker-entrypoint.…"   3 months ago     Exited (255) 20 hours ago   0.0.0.0:9091->80/tcp                 thirsty_pare
fe399986e504   9d0e6f6199dc         "/docker-entrypoint.…"   3 months ago     Exited (255) 20 hours ago   0.0.0.0:9090->80/tcp                 infallible_germain
5be8ea3f3bb2   9d0e6f6199dc         "/docker-entrypoint.…"   3 months ago     Created                                                          ecstatic_jennings
1e61b2cb20bf   hello-world          "/hello"                 3 months ago     Exited (0) 3 months ago                                          nervous_elbakyan
23106efee7f4   9d0e6f6199dc         "/docker-entrypoint.…"   8 months ago     Exited (255) 4 months ago   0.0.0.0:8081->80/tcp                 my-nginx
d1c21cfdc92b   jenkins/jenkins:lts  "/usr/bin/tini -- /u…"   16 months ago    Exited (255) 8 months ago   50000/tcp, 0.0.0.0:9090->8080/tcp    jenkins
6d895ae2b5eb   d1397258b209         "docker-entrypoint.s…"   2 years ago      Exited (0) 2 years ago                                           nice_greider
1f99a9cc77bf   gradle               "/__cacert_entrypoin…"   2 years ago      Exited (0) 2 years ago                                           elastic_stonebraker
77e3fc5c920f   node                 "docker-entrypoint.s…"   2 years ago      Exited (0) 2 years ago                                           con234
7b58e9788533   node                 "docker-entrypoint.s…"   2 years ago      Exited (0) 2 years ago                                           containerblah
0dc3412adadd   postgres             "docker-entrypoint.s…"   2 years ago      Exited (1) 2 years ago                                           stoic_mclaren
(base) PS C:\Users\203> docker logs aryank-container
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://172.17.0.2:5000
Press CTRL+C to quit
172.17.0.1 - - [21/Jul/2026 05:39:56] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [21/Jul/2026 05:39:56] "GET /favicon.ico HTTP/1.1" 404 -
```

```
C:\Users\203\Desktop\docker-project>docker images
REPOSITORY         TAG          IMAGE ID       CREATED          SIZE
aryank             latest       cd0827da2eff   7 seconds ago    132MB
<none>             <none>       42f468bdae9d   3 minutes ago    132MB
nginx              latest       4e5db4761e0f   5 days ago       161MB
python             3.12-slim    25c5b8011a34   7 days ago       119MB
ubuntu             latest       de7345b16e94   7 days ago       100MB
my-nginx           latest       c18468aa2adf   3 months ago     152MB
hello-world        latest       e2ac70e7319a   3 months ago     10.1kB
nginx              <none>       9d0e6f6199dc   8 months ago     152MB
jenkins/jenkins    lts          6a44d1dd2d60   16 months ago    468MB
docker             dind         a16883c00816   16 months ago    395MB
node               latest       8bbb9d1bc633   2 years ago      1.1GB
sample-web-app     latest       d03cd9229681   2 years ago      912MB
gradle             latest       716c875ae781   2 years ago      713MB
postgres           latest       eb634efa7ee4   2 years ago      431MB
redis              latest       170a1e90f843   2 years ago      138MB
redis              <none>       d1397258b209   2 years ago      138MB

C:\Users\203\Desktop\docker-project>docker run --name aryank-container -p 5000:5000 aryank
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://172.17.0.2:5000
Press CTRL+C to quit
172.17.0.1 - - [21/Jul/2026 05:39:56] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [21/Jul/2026 05:39:56] "GET /favicon.ico HTTP/1.1" 404 -

C:\Users\203\Desktop\docker-project>
```

```
(base) PS C:\Users\203> docker logs aryank-container
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://172.17.0.2:5000
Press CTRL+C to quit
172.17.0.1 - - [21/Jul/2026 05:39:56] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [21/Jul/2026 05:39:56] "GET /favicon.ico HTTP/1.1" 404 -
(base) PS C:\Users\203> docker inspect aryank-container
[
    {
        "Id": "103ab71a72254f273a9076780346f04e0e1e4e6c5eff84f2de0302b0be97b20c",
        "Created": "2026-07-21T05:39:07.077385271Z",
        "Path": "python",
        "Args": [
            "app.py"
        ],
        "State": {
            "Status": "running",
            "Running": true,
            "Paused": false,
            "Restarting": false,
            "OOMKilled": false,
            "Dead": false,
            "Pid": 1921,
            "ExitCode": 0,
            "Error": "",
            "StartedAt": "2026-07-21T05:39:07.458740369Z",
            "FinishedAt": "0001-01-01T00:00:00Z"
        },
```

```
(base) PS C:\Users\203> docker ps
CONTAINER ID   IMAGE     COMMAND           CREATED         STATUS         PORTS                    NAMES
103ab71a7225   aryank    "python app.py"   4 minutes ago   Up 4 minutes   0.0.0.0:5000->5000/tcp   aryank-container
(base) PS C:\Users\203> docker exec -it aryank-container sh
# ls
app.py  requirements.txt
# exit
(base) PS C:\Users\203> docker run -d --name aryank-background -p 5001:5000 aryank
3aa03439b164f2351283c4216937b82c57fb27ea38be1f3e2fd9b6b56c3921e5
(base) PS C:\Users\203> docker ps
CONTAINER ID   IMAGE     COMMAND           CREATED         STATUS         PORTS                    NAMES
3aa03439b164   aryank    "python app.py"   5 seconds ago   Up 5 seconds   0.0.0.0:5001->5000/tcp   aryank-background
103ab71a7225   aryank    "python app.py"   8 minutes ago   Up 8 minutes   0.0.0.0:5000->5000/tcp   aryank-container
(base) PS C:\Users\203> docker ps -a
CONTAINER ID   IMAGE                COMMAND                  CREATED          STATUS                          PORTS                                      NAMES
3aa03439b164   aryank               "python app.py"          12 seconds ago   Up 12 seconds                   0.0.0.0:5001->5000/tcp                     aryank-background
103ab71a7225   aryank               "python app.py"          8 minutes ago    Up 8 minutes                    0.0.0.0:5000->5000/tcp                     aryank-container
9b433640960c   ubuntu               "/bin/bash"              54 minutes ago   Exited (0) 48 minutes ago                                                  eager_bartik
79d59be7b7ef   ubuntu               "/bin/bash"              55 minutes ago   Exited (0) 55 minutes ago                                                  quirky_matsumoto
c8fe1146dd791   ubuntu               "bash"                   20 hours ago     Exited (255) About an hour ago                                             pedantic_hypatia
2ba1c97abb52   ubuntu               "/bin/bash"              20 hours ago     Exited (127) 20 hours ago                                                  suspicious_hugle
0fb51210958c   my-nginx             "/docker-entrypoint.…"   3 months ago     Exited (255) 20 hours ago       0.0.0.0:9091->80/tcp                       thirsty_pare
fe399986e504   9d0e6f6199dc         "/docker-entrypoint.…"   3 months ago     Exited (255) 20 hours ago       0.0.0.0:9090->80/tcp                       infallible_germain
5be8ea3f3bb2   9d0e6f6199dc         "/docker-entrypoint.…"   3 months ago     Created                                                                    ecstatic_jennings
1e61b2cb20bf   hello-world          "/hello"                 3 months ago     Exited (0) 3 months ago                                                    nervous_elbakyan
23106efee7f4   9d0e6f6199dc         "/docker-entrypoint.…"   8 months ago     Exited (255) 4 months ago       0.0.0.0:8081->80/tcp                       my-nginx
d1c21cfdc92b   jenkins/jenkins:lts  "/usr/bin/tini -- /u…"   16 months ago    Exited (255) 8 months ago       50000/tcp, 0.0.0.0:9090->8080/tcp          jenkins
6d895ae2b5eb   d1397258b209         "docker-entrypoint.s…"   2 years ago      Exited (0) 2 years ago                                                     nice_greider
1f99a9cc77bf   gradle               "/__cacert_entrypoin…"   2 years ago      Exited (0) 2 years ago                                                     elastic_stonebraker
77e3fc5c920f   node                 "docker-entrypoint.s…"   2 years ago      Exited (0) 2 years ago                                                     con234
7b58e9788533   node                 "docker-entrypoint.s…"   2 years ago      Exited (0) 2 years ago                                                     containerblah
0dc3412adadd   postgres             "docker-entrypoint.s…"   2 years ago      Exited (1) 2 years ago                                                     stoic_mclaren
(base) PS C:\Users\203> docker stop aryank-container
aryank-container
(base) PS C:\Users\203> docker start aryank-container
aryank-container
(base) PS C:\Users\203> docker rm aryank-container
Error response from daemon: cannot remove container "/aryank-container": container is running: stop the container before removing or force remove
(base) PS C:\Users\203> docker stop aryank-container
aryank-container
(base) PS C:\Users\203> docker rm aryank-container
aryank-container
```

```
(base) PS C:\Users\203> docker ps -a
CONTAINER ID   IMAGE                COMMAND                  CREATED             STATUS                          PORTS                                  NAMES
3aa03439b164   aryank               "python app.py"          About a minute ago  Up About a minute               0.0.0.0:5001->5000/tcp                 aryank-background
9b433640960c   ubuntu               "/bin/bash"              56 minutes ago      Exited (0) 49 minutes ago                                              eager_bartik
79d59be7b7ef   ubuntu               "/bin/bash"              57 minutes ago      Exited (0) 57 minutes ago                                              quirky_matsumoto
c8fe146dd791   ubuntu               "bash"                   20 hours ago        Exited (255) About an hour ago                                         pedantic_hypatia
2ba1c97abb52   ubuntu               "/bin/bash"              20 hours ago        Exited (127) 20 hours ago                                              suspicious_hugle
0fb51210958c   my-nginx             "/docker-entrypoint.…"   3 months ago        Exited (255) 20 hours ago       0.0.0.0:9091->80/tcp                   thirsty_pare
fe399986e504   9d0e6f6199dc         "/docker-entrypoint.…"   3 months ago        Exited (255) 20 hours ago       0.0.0.0:9090->80/tcp                   infallible_germain
5be8ea3f3bb2   9d0e6f6199dc         "/docker-entrypoint.…"   3 months ago        Created                                                                ecstatic_jennings
1e61b2cb20bf   hello-world          "/hello"                 3 months ago        Exited (0) 3 months ago                                                nervous_elbakyan
23106efee7f4   9d0e6f6199dc         "/docker-entrypoint.…"   8 months ago        Exited (255) 4 months ago       0.0.0.0:8081->80/tcp                   my-nginx
d1c21cfdc92b   jenkins/jenkins:lts  "/usr/bin/tini -- /u…"   16 months ago       Exited (255) 8 months ago       50000/tcp, 0.0.0.0:9090->8080/tcp      jenkins
6d895ae2b5eb   d1397258b209         "docker-entrypoint.s…"   2 years ago         Exited (0) 2 years ago                                                 nice_greider
1f99a9cc77bf   gradle               "/__cacert_entrypoin…"   2 years ago         Exited (0) 2 years ago                                                 elastic_stonebraker
77e3fc5c920f   node                 "docker-entrypoint.s…"   2 years ago         Exited (0) 2 years ago                                                 con234
7b58e9788533   node                 "docker-entrypoint.s…"   2 years ago         Exited (0) 2 years ago                                                 containerblah
0dc3412adadd   postgres             "docker-entrypoint.s…"   2 years ago         Exited (1) 2 years ago                                                 stoic_mclaren
(base) PS C:\Users\203> docker logs -f aryank-background
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://172.17.0.3:5000
Press CTRL+C to quit
(base) PS C:\Users\203> docker inspect aryank-background
[
    {
        "Id": "3aa03439b164f2351283c4216937b82c57fb27ea38be1f3e2fd9b6b56c3921e5",
        "Created": "2026-07-21T05:47:22.170027047Z",
        "Path": "python",
        "Args": [
            "app.py"
        ],
        "State": {
            "Status": "running",
            "Running": true,
```

```
(base) PS C:\Users\203> docker container stats
CONTAINER ID   NAME                CPU %     MEM USAGE / LIMIT    MEM %     NET I/O     BLOCK I/O   PIDS
3aa03439b164   aryank-background   0.00%     21.23MiB / 15.5GiB   0.13%     866B / 0B   0B / 0B     1
CONTAINER ID   NAME                CPU %     MEM USAGE / LIMIT    MEM %     NET I/O     BLOCK I/O   PIDS
3aa03439b164   aryank-background   0.00%     21.23MiB / 15.5GiB   0.13%     866B / 0B   0B / 0B     1
CONTAINER ID   NAME                CPU %     MEM USAGE / LIMIT    MEM %     NET I/O     BLOCK I/O   PIDS
3aa03439b164   aryank-background   0.01%     21.23MiB / 15.5GiB   0.13%     866B / 0B   0B / 0B     1
CONTAINER ID   NAME                CPU %     MEM USAGE / LIMIT    MEM %     NET I/O     BLOCK I/O   PIDS
3aa03439b164   aryank-background   0.01%     21.23MiB / 15.5GiB   0.13%     866B / 0B   0B / 0B     1
CONTAINER ID   NAME                CPU %     MEM USAGE / LIMIT    MEM %     NET I/O     BLOCK I/O   PIDS
3aa03439b164   aryank-background   0.01%     21.23MiB / 15.5GiB   0.13%     866B / 0B   0B / 0B     1
CONTAINER ID   NAME                CPU %     MEM USAGE / LIMIT    MEM %     NET I/O     BLOCK I/O   PIDS
3aa03439b164   aryank-background   0.01%     21.23MiB / 15.5GiB   0.13%     866B / 0B   0B / 0B     1
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CONTAINER ID   NAME                CPU %     MEM USAGE / LIMIT    MEM %     NET I/O     BLOCK I/O   PIDS
3aa03439b164   aryank-background   0.01%     21.23MiB / 15.5GiB   0.13%     866B / 0B   0B / 0B     1
CONTAINER ID   NAME                CPU %     MEM USAGE / LIMIT    MEM %     NET I/O     BLOCK I/O   PIDS
3aa03439b164   aryank-background   0.01%     21.23MiB / 15.5GiB   0.13%     866B / 0B   0B / 0B     1
CONTAINER ID   NAME                CPU %     MEM USAGE / LIMIT    MEM %     NET I/O     BLOCK I/O   PIDS
3aa03439b164   aryank-background   0.01%     21.23MiB / 15.5GiB   0.13%     866B / 0B   0B / 0B     1
CONTAINER ID   NAME                CPU %     MEM USAGE / LIMIT    MEM %     NET I/O     BLOCK I/O   PIDS
3aa03439b164   aryank-background   0.01%     21.23MiB / 15.5GiB   0.13%     866B / 0B   0B / 0B     1
CONTAINER ID   NAME                CPU %     MEM USAGE / LIMIT    MEM %     NET I/O     BLOCK I/O   PIDS
3aa03439b164   aryank-background   0.01%     21.23MiB / 15.5GiB   0.13%     866B / 0B   0B / 0B     1
CONTAINER ID   NAME                CPU %     MEM USAGE / LIMIT    MEM %     NET I/O     BLOCK I/O   PIDS
3aa03439b164   aryank-background   0.01%     21.23MiB / 15.5GiB   0.13%     866B / 0B   0B / 0B     1
```

```
(base) PS C:\Users\203> docker search python
NAME                      DESCRIPTION                                     STARS     OFFICIAL
python                    Python is an interpreted, interactive, objec…   10490     [OK]
circleci/python           Python is an interpreted, interactive, objec…   116
cimg/python                                                               25
demisto/python            The Demisto base python image                   5
ubuntu/python             A chiselled Ubuntu rock with the Python runt…   33
okteto/python                                                             0
paketobuildpacks/python                                                   0
cleanstart/python         Deploy, scale, and secure critical infrastru…   1
activestate/python        ActiveState's customizable, low-to-no vulner…   11
intel/python                                                              5
corpusops/python          https://github.com/corpusops/docker-images/     1
rootpublic/python                                                         0
gentkit/python            A Docker images of Python environment .         0
stagex/python             High-level scripting language                   0
linuxserver/python                                                        0
vulhub/python                                                             2
koxudaxi/python           Dockerfiles for CPython of lysnikolaou's tag…   0
unidata/python            Container with Unidata's python packages as …   3
clarinpl/python                                                           0
dtcenter/python           Necessary Python3 packages for plotting GRIB…   1
openeuler/python                                                          1
dragonflyoss/python                                                       0
nanozoo/python                                                            0
hkube/python                                                              0
python/docker-bpo         Docker image for local development of http:/…   0
(base) PS C:\Users\203> docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
Digest: sha256:5a88c9c45479443d7be2eadc894b4ed0a9801bae03d97a5760ae13b5c2005942
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest
```

CONCLUSION :

The experiment successfully demonstrated the basic use of Docker for containerizing and running a Python Flask application. We created a Docker image using a Dockerfile, ran and managed containers, viewed logs, inspected containers, and practiced essential Docker commands for images, containers, and Docker Hub.